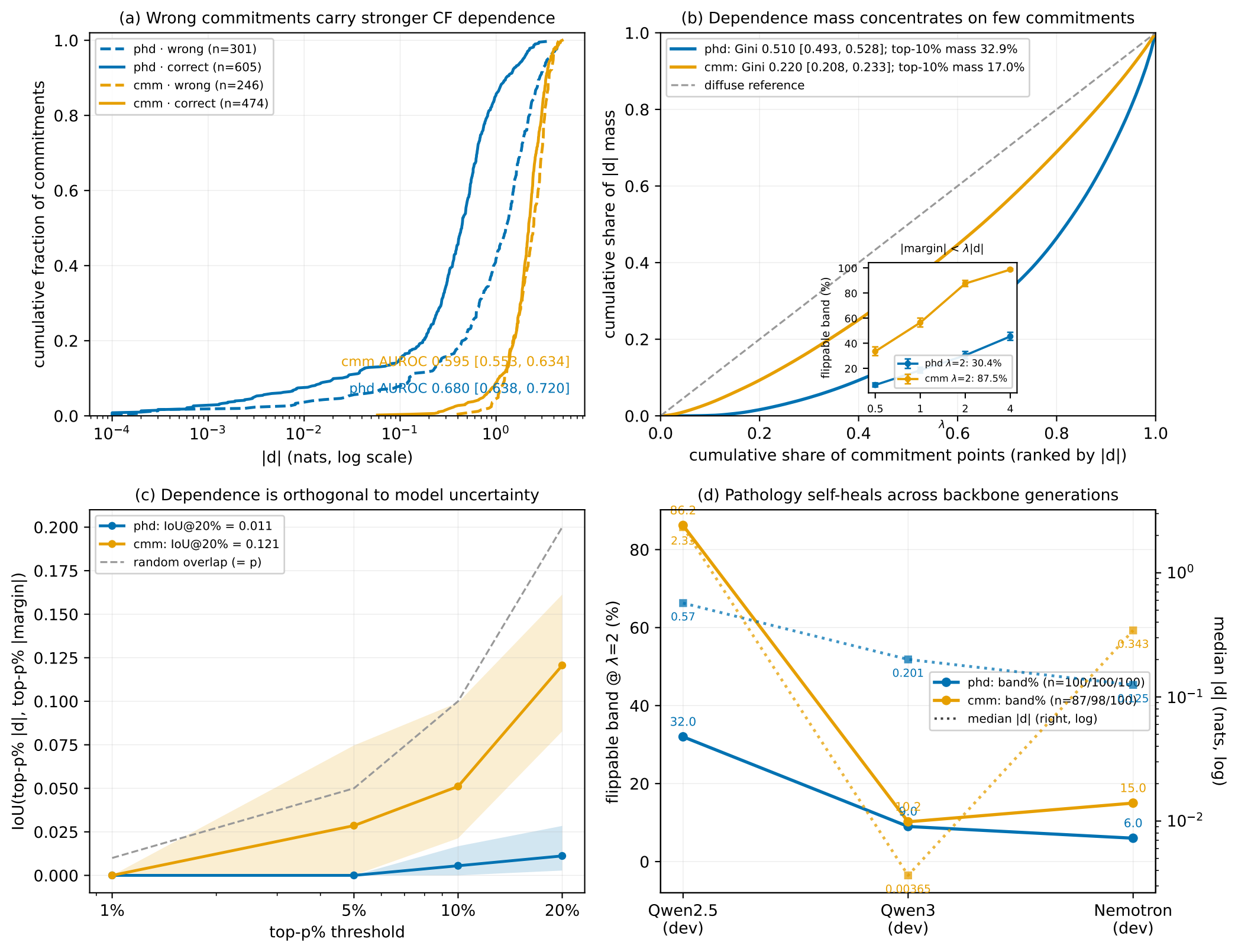
(a) Wrong commitments carry stronger CF dependence
phd · wrong (n=301)
phd · correct (n=605)
cmm · wrong (n=246)
cmm · correct (n=474)
cmm AUROC 0.595 [0.553, 0.634]
phd AUROC 0.680 [0.638, 0.720]
cumulative fraction of commitments
|d| (nats, log scale)
(b) Dependence mass concentrates on few commitments
phd: Gini 0.510 [0.493, 0.528]; top-10% mass 32.9%
cmm: Gini 0.220 [0.208, 0.233]; top-10% mass 17.0%
diffuse reference
cumulative share of |d| mass
cumulative share of commitment points (ranked by |d|)
|margin| < λ|d|
flippable band (%)
phd λ=2: 30.4%
cmm λ=2: 87.5%
(c) Dependence is orthogonal to model uncertainty
phd: IoU@20% = 0.011
cmm: IoU@20% = 0.121
random overlap (= p)
IoU(top-p% |d|, top-p% |margin|)
top-p% threshold
(d) Pathology self-heals across backbone generations
flippable band @ λ=2 (%)
median |d| (nats, log)
phd: band% (n=100/100/100)
cmm: band% (n=87/98/100)
median |d| (right, log)
Qwen2.5 (dev)
Qwen3 (dev)
Nemotron (dev)